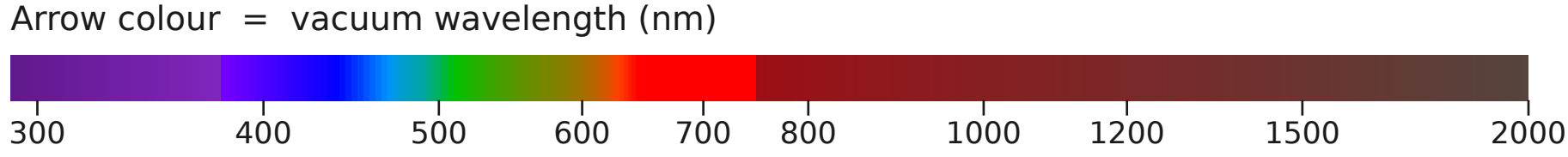


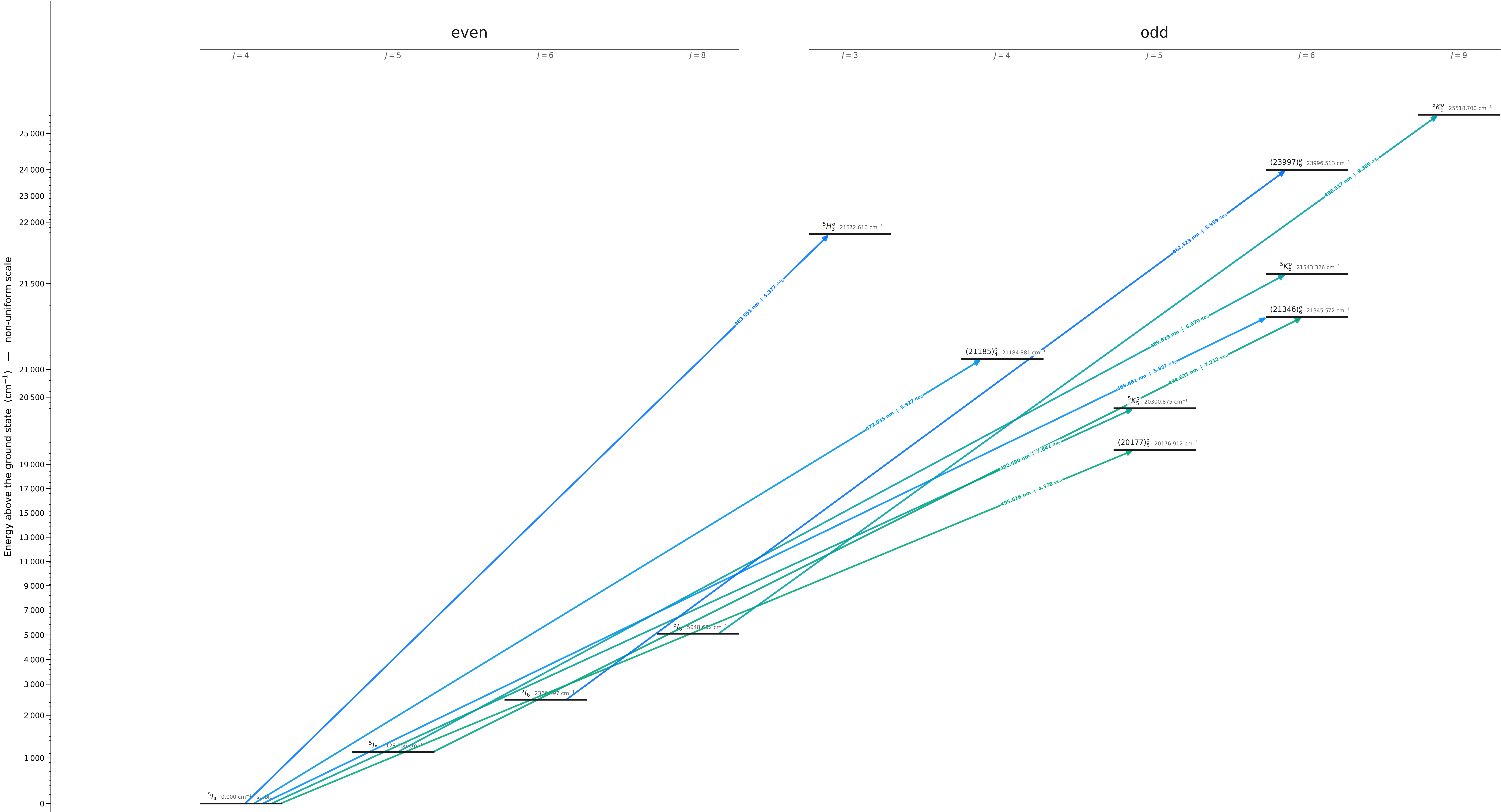
# Neodymium

Arrow label: vacuum wavelength | reduced dipole matrix element  $|(U||e r||J')|$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Nd (Nd I). Strongest classified transitions below 2  $\mu\text{m}$  (NIST): 12 levels, 9 lines.

Nd<sup>+</sup> ionisation limit 44560.11  $\text{cm}^{-1}$  = 5.52475 eV (224.416 nm)



## Strongest transitions

| Transition            | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $ (U  e r  J') $ ( $e a_0$ ) | A ( $\text{s}^{-1}$ ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use |
|-----------------------|--------------------|--------------------|-------------|------------------------------|-----------------------|-------------------------|------------|-----|
| $^5I_6 - (23997)^5_6$ | 462.3226           | 462.1932           | 648.4486    | 5.959                        | 5.600e+07             | -                       | -          | -   |
| $^5I_4 - ^5H_3^o$     | 463.5508           | 463.4210           | 646.7306    | 5.377                        | 8.400e+07             | -                       | -          | -   |
| $^5I_4 - (21346)^5_6$ | 468.4812           | 468.3501           | 639.9242    | 5.857                        | 5.200e+07             | -                       | -          | -   |
| $^5I_4 - (21185)^5_4$ | 472.0347           | 471.9027           | 635.1068    | 3.927                        | 3.300e+07             | -                       | -          | -   |
| $^5I_8 - ^5K_9^o$     | 488.5174           | 488.3810           | 613.6781    | 9.809                        | 8.800e+07             | -                       | -          | -   |
| $^5I_5 - ^5K_8^o$     | 489.8294           | 489.6927           | 612.0344    | 6.670                        | 5.900e+07             | -                       | -          | -   |
| $^5I_4 - ^5K_5^o$     | 492.5896           | 492.4521           | 608.6049    | 7.642                        | 9.000e+07             | -                       | -          | -   |
| $^5I_5 - (21346)^5_6$ | 494.6206           | 494.4826           | 606.1059    | 7.212                        | 6.700e+07             | -                       | -          | -   |
| $^5I_4 - (20177)^5_5$ | 495.6160           | 495.4777           | 604.8886    | 4.378                        | 2.900e+07             | -                       | -          | -   |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
 $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

## <sup>143</sup>Nd hyperfine structure ( $I = 7/2$ )

| Level   | A (MHz)     | B (MHz)    | Shift of each F level from the centre of gravity (MHz) |                 |                 |                 |                 |                |                                   |
|---------|-------------|------------|--------------------------------------------------------|-----------------|-----------------|-----------------|-----------------|----------------|-----------------------------------|
| $^5I_4$ | -195.649(9) | 122.25(28) | F=1/2: +3483.89                                        | F=3/2: +3174.98 | F=5/2: +2663.25 | F=7/2: +1953.37 | F=9/2: +1051.90 | F=11/2: -32.75 | F=13/2: -1290.27 F=15/2: -2708.52 |

Measured A, B constants; sources in the data file.